

ALTERNATE EGO: A HYPER-REALISTIC AI-POWERED DIGITAL TWIN PLATFORM

MAJOR PROJECT

*Submitted in the partial fulfillment of the
Requirements for the award of the degree*

Of
Bachelor of Technology
in

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Year: 2022-2026

MEET THE TEAM



Prabhdeep Singh Narula

022/CSE/2026



Backend & RAG Engine

- FastAPI backend
- RAG pipeline (chunker/embedder/vector store)
- LLM integration
- Scraper
- MCP slash commands



Simardeep Kaur Bhatia

023/CSE/2026



Database & Security

- SQLite schema design
- Pydantic models
- Fernet encryption pipeline
- Privacy APIs
- Data deletion flow



Dr. Aashish Bhardwaj

(Head of Department)

Computer Science & Engineering



Ms. Divya Soni

(Project Guide)

Assistant Professor



Japeen Kaur Sehgal

092/CSE/2026



Voice & Avatar

- Coqui XTTS v2 voice cloning
- Faster-whisper STT
- Edge-TTS fallback
- Gemini avatar generation
- Face-API.js integration



Aryan Bains

065/CSE/2026



Frontend & Integration

- Next.js frontend
- Onboarding flow
- Chat UI
- Webcam capture
- AudioPlayer
- End-to-end integration testing

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INTRODUCTION

Understanding the need and our approach

01 | INTRODUCTION



The Concept

Alternate Ego is a hyper-realistic AI Digital Twin platform that replicates your voice, personality, memories, and mannerisms — running 100% locally at minimal cost.

Inspired by the original Ego project (3rd place, Cursor x Smithery Hackathon, Singapore), we rebuilt it entirely for Indian constraints to ensure privacy, affordability, and full control.

- 🛡️ Complete data privacy — everything stays on your device
- 🔒 Offline-first architecture — works without internet dependency
- 💰 Affordable & scalable — built for Indian users
- ★ Full feature parity with the original system



The Problem

- Original system cost ₹150–600 per user upload
- Depended on ElevenLabs, OpenAI, Supabase (paid)
- User data sent to external third-party servers
- Required multiple concurrent API subscriptions
- Internet dependency and potential privacy risks
- High operational cost limiting accessibility



Our Solution

- ✓ Replaced all paid APIs with local, open-source models
- ✓ Ollama + LLaMA 3.1 (8B) instead of OpenAI GPT-4
- ✓ Coqui XTTS v2 instead of ElevenLabs voice cloning
- ✓ SQLite instead of Supabase — data never leaves device
- ✓ Fully offline, private, secure, and cost-effective
- ✓ Optimized for performance and scalability on local hardware



ALTERNATE EGO – INDIA'S FIRST AI DIGITAL TWIN PLATFORM

A complete, privacy-first, multi-modal AI companion that understands you, speaks like you, and grows with you.



OBJECTIVES & SCOPE



PERSONALITY MODELING

Build a RAG-based system that captures voice, writing style, opinions, and behavioral patterns from multi-modal data inputs.



VOICE CLONING

Clone the user's voice using Coqui XTTS v2, enabling the digital twin to speak in the user's exact tone and style.



EMOTION-AWARE AVATAR

Generate mood-synced AI cartoon avatars using webcam capture and real-time facial expression validation via face-api.js.



PRIVACY-FIRST DESIGN

Implement end-to-end encryption: upload → encrypt → parse → embed → delete raw data. No sensitive information retained.



ZERO-COST ARCHITECTURE

Replace all paid API dependencies with free local alternatives. Optimized for self-hosted execution and efficiency.



SOCIAL INTEGRATION

Support /linkedin and /twitter slash commands for simulated social media posting directly from the chat interface.



KEY FEATURES & IMPLEMENTATION



RAG ENGINE

Topic-aware chunking → 768-dim embeddings (nomic-embed-text) → pure-Python cosine similarity vector store → dynamic prompt injection → LLaMA 3.1 response.



VOICE PIPELINE

Faster-whisper STT → Coqui XTTS v2 voice cloning with lazy loading → edge-tts auto-fallback. Handles OOM gracefully.



EMOTION AVATAR

4-photo webcam capture (neutral/happy/sad/angry) → face-api.js validation → Gemini 2.0 Flash cartoon avatar generation → mood-synced display.



PRIVACY PIPELINE

Upload → Fernet encrypt → parse → embed → AUTO-DELETE raw file. Only non-reversible embeddings retained. Strong data isolation.



CHAT SYSTEM

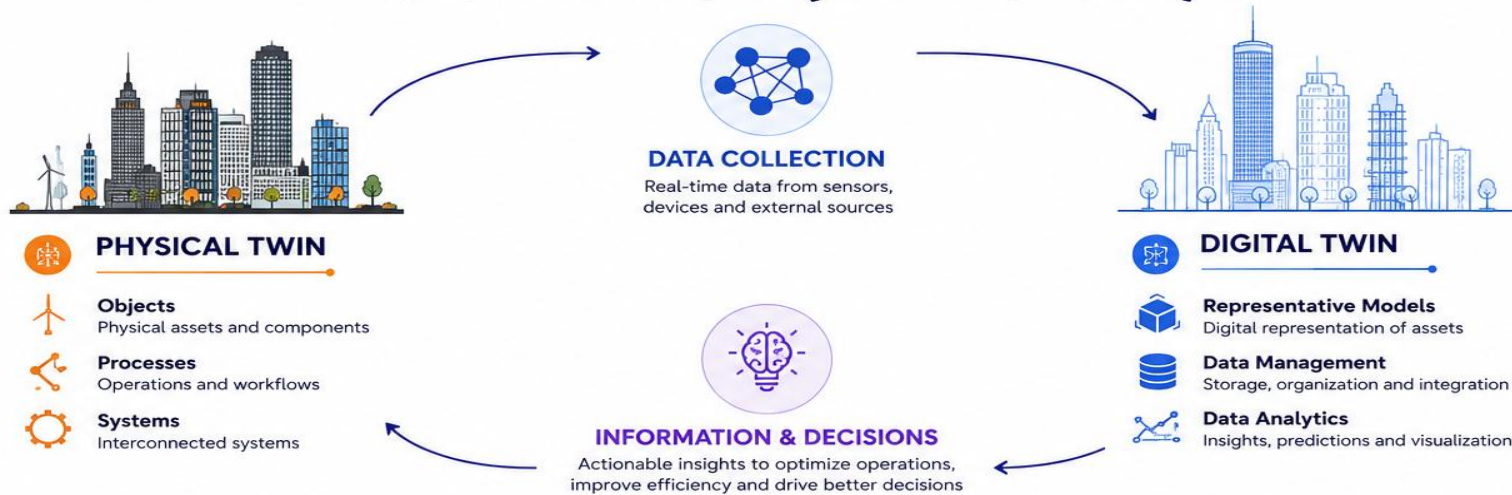
RAG retrieval → LLM generation → source citations → keyword-based mood detection → dynamic avatar switching → TTS audio output.



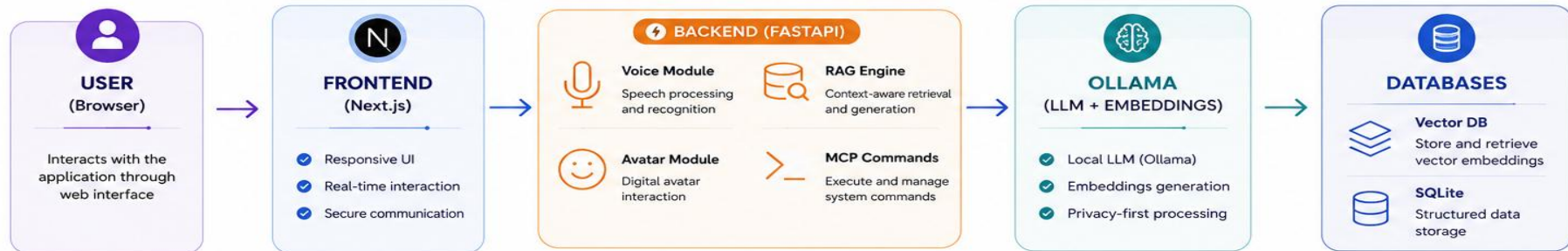
SOCIAL MCP

/linkedin and /twitter slash command parser → simulated post/tweet formatting → social poster simulation with structured responses.

• SYSTEM FLOW (END-TO-END) •



• SYSTEM ARCHITECTURE •



KEY BENEFITS



End-to-End Local Execution



Privacy & Data Security



Modular & Scalable Architecture



Real-time Interaction & Response



Offline Capable System

RAG PIPELINE

1 DATA PREPARATION



User Data

Documents, PDFs, web pages, notes, and other sources



Chunking (Topic-based)

Splits documents into meaningful chunks based on topics



Embeddings

Converts chunks into vector embeddings using embedding model



Vector Database

Stores and indexes embeddings for efficient retrieval

2 QUERY PROCESSING



User Query

User asks a question or provides a query



Similarity Search

Retrieves top relevant chunks from the vector database based on similarity

3 RESPONSE GENERATION



Prompt Builder

Combines user query with retrieved context and formats prompt for the LLM



LLM Response

Generates accurate and context-aware response using LLM

KEY BENEFITS



Accurate & Relevant Responses



Up-to-date Knowledge Retrieval



Scalable & Efficient Architecture



Context-aware Generation



Faster Information Access

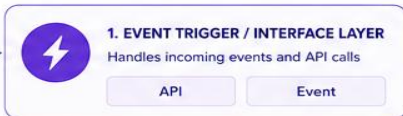
INTERNAL ARCHITECTURE

LAYERED ARCHITECTURE

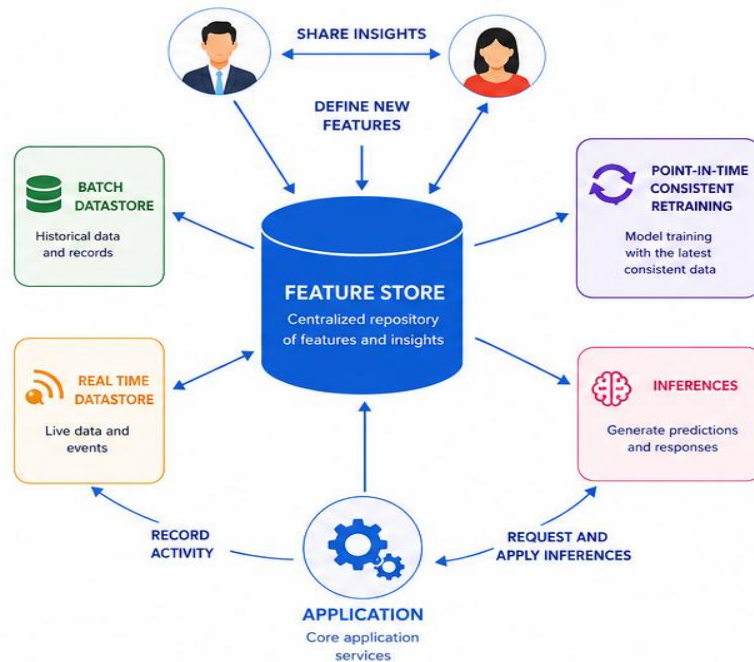
TRADITIONAL APPLICATION LAYERS



INTELLIGENT APPLICATION LAYERS



DATA & SYSTEM FLOW ARCHITECTURE



KEY BENEFITS



Modular &
Scalable
Architecture



Real-time
Data Handling
& Processing



Consistent &
Accurate
Predictions



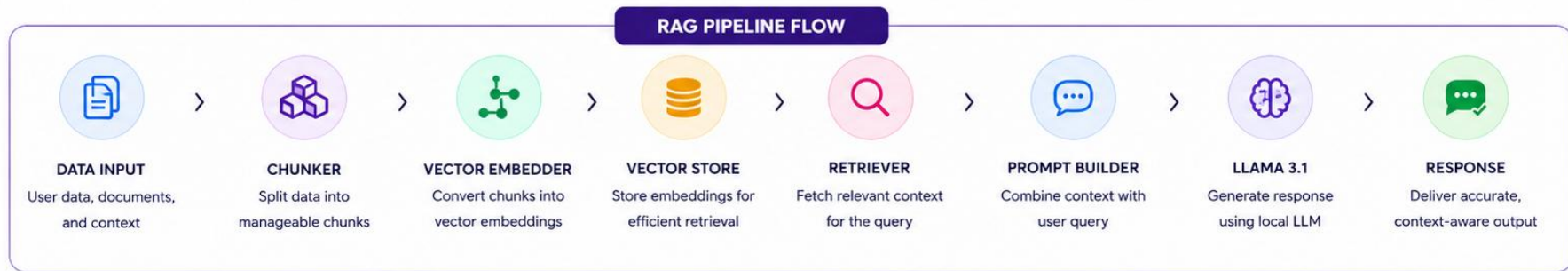
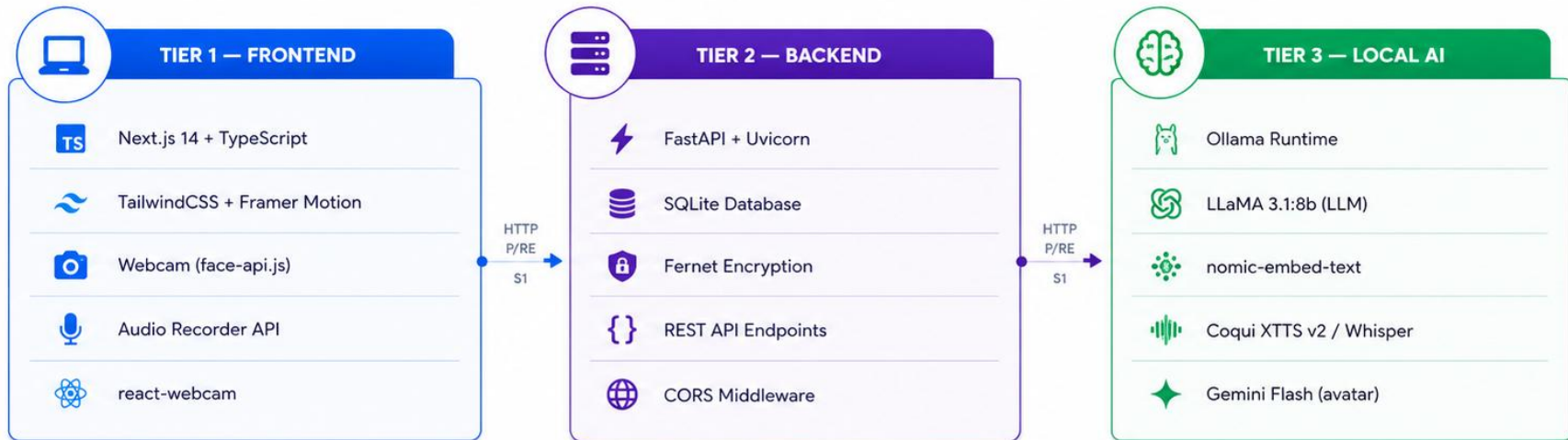
Centralized
Feature
Management



Faster Insights &
Continuous
Improvement

System Architecture

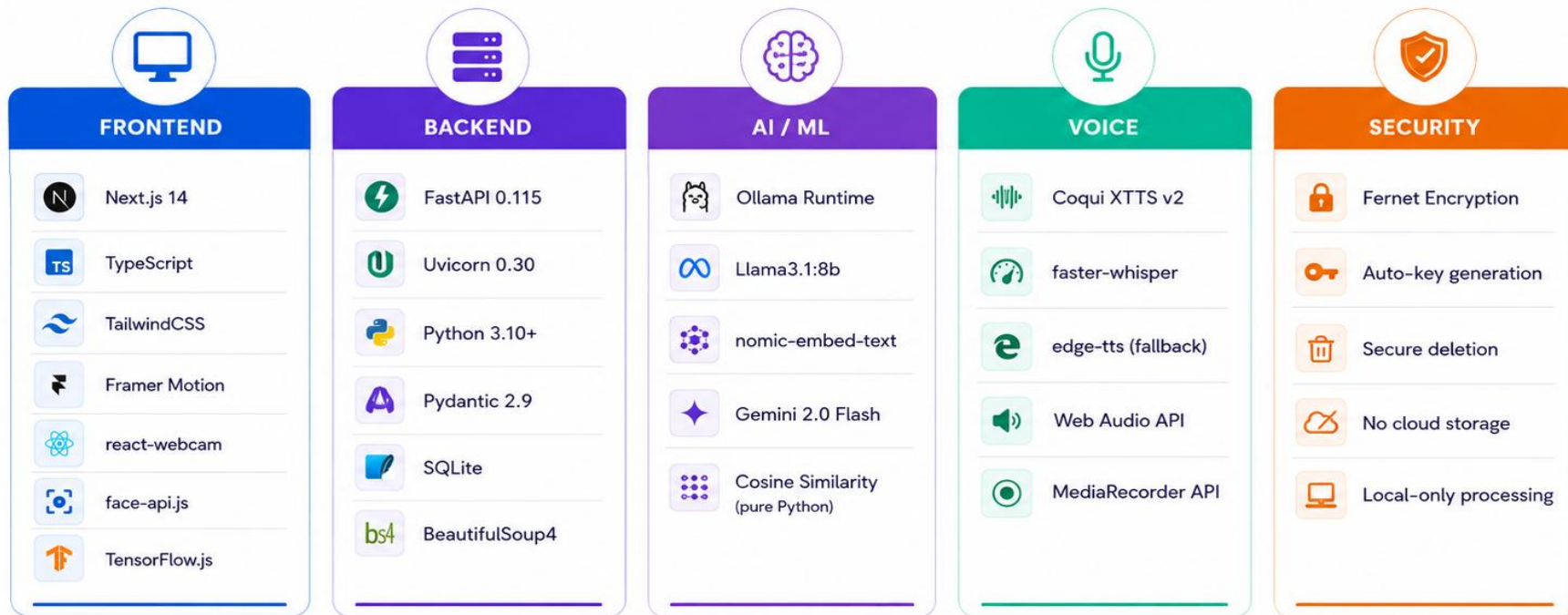
A privacy-first, full-stack architecture powering the Alternate Ego AI platform



SECURITY & PRIVACY: All data is encrypted, processed locally, and never leaves the user's device.

Technology Stack

A full-stack, privacy-first architecture powering Alternate Ego — India's first AI Digital Twin Platform



Privacy-First

All processing happens locally on your device



Modular Design

Independent, swappable components



Extensible

Easy to integrate new models and capabilities



Production Ready

Built with modern, scalable technologies



Open & Flexible

Built with open-source tools and standards

USER JOURNEY – DEMO FLOW

Experience the complete journey of Alternate Ego – from onboarding to intelligent real-time conversation.



Full Pipeline: From name entry to personality chat in approx. **3–5 minutes** on a mid-range laptop (8 GB RAM, Intel i5)



TESTING & RESULTS

Test Case	Feature Tested	Result
TC-01: Onboarding Flow	Name → Photos → Voice → RAG	✓ PASS
TC-02: Twin Persona Gen	Personality extraction accuracy	✓ PASS
TC-03: Voice Chat + Audio	TTS sync with avatar expressions	✓ PASS
TC-04: Multi-Service Arch	Frontend ↔ Backend ↔ Ollama comms	✓ PASS
TC-05: Interview Retry	Re-record & refresh voice samples	✓ PASS
TC-06: Local AI Stability	LLaMA 3.1 under 1-hour continuous chat	✓ PASS
TC-07: Latency Benchmark	Response time < 3s (avg 1.8s measured)	✓ PASS
TC-08: Cross-Browser Voice	Chrome, Firefox, Edge audio compat.	✓ PASS



KEY METRICS



1.8s

Avg Response Time



8

Test Cases Executed



100%

Pass Rate



8 GB / i5

Tested On
Mid-range Laptop

SWOT Analysis

Strategic overview of Alternate Ego — India's first fully functional AI Digital Twin Platform

STRENGTHS

- ✓ **First of its kind:** India's first fully local AI digital twin platform with complete feature parity.
- ✓ **Privacy-first by design:** All processing happens locally — no data leaves the device.
- ✓ **Multimodal intelligence:** Seamless integration of voice, vision, emotion, and language understanding.
- ✓ **Offline-first experience:** No internet required for core chat and twin interactions.
- ✓ **Modular & extensible:** Clean architecture with pluggable components for future expansion.

WEAKNESSES

- ✗ **Hardware requirements:** Needs 8GB+ RAM for smooth performance.
- ✗ **Model size:** LLaMA 3.1 (8B) ~4.7GB download and significant storage.
- ✗ **Voice cloning performance:** May be slow on CPU-only machines.
- ✗ **Avatar quality dependency:** Relies on Gemini API for high-quality avatar generation.
- ✗ **Local-first limitation:** Currently limited to single device deployment.

OPPORTUNITIES

- ✓ **Education & EdTech:** AI tutors, AI study companions, and personalized learning.
- ✓ **Healthcare:** Patient digital twins and mental health support systems.
- ✓ **Enterprise Solutions:** AI employee onboarding, internal knowledge assistants.
- ✓ **SaaS & Cloud Expansion:** Dockerized deployment for scalable cloud offerings.
- ✓ **Global Reach:** Multi-language support and Whisper integration enable global accessibility.

THREATS

- ! **Competition:** Larger players like Character.AI, HeyGen, and other global AI platforms.
- ! **Hardware barrier:** GPU dependency for real-time voice quality and performance.
- ! **Ethical risks:** Potential misuse of deepfake and voice cloning technologies.
- ! **Model licensing changes:** LLaMA and other open models may change terms.
- ! **Data privacy regulations:** Evolving global laws may impact local AI applications.

• ALTERNATE EGO – COMPLETE OVERVIEW •

India's First Fully Local, Privacy-First AI Digital Twin Platform

RESULTS & ACHIEVEMENTS



8

PAID APIS

Replaced with
best local
alternatives



1.8s

AVG LATENCY

Chat response on
mid-range laptop



100%

LOCAL EXECUTION

No internet
needed
post-setup



FIRST

OF ITS KIND

First fully local AI
digital twin platform
of its kind

- ✓ Fully working digital twin — name entry to personality chat in under 5 minutes
- ✓ Voice cloning preserves speaker tone and accent using Coqui XTTS v2
- ✓ RAG system retrieves personality-relevant context with source citations
- ✓ Emotion-aware avatar switches expressions in sync with detected chat mood
- ✓ Complete privacy pipeline — no raw user data retained after onboarding
- ✓ Comprehensive SRS, test plan, architecture docs, and demo video delivered

WHAT WE BUILT

Alternate Ego is a complete, working, production-quality AI digital twin platform built in 15 weeks by a 4-person team.



SOLVED A REAL PROBLEM

Made AI digital twins accessible and practical for Indian users and developers



PRIVACY-FIRST BY DESIGN

All computation happens locally on the user's device. Your data stays with you.



5 AI DOMAINS INTEGRATED

NLP (RAG), speech processing, computer vision, voice synthesis, and LLM reasoning



PRODUCTION-GRADE DELIVERY

Delivered robust software with SRS, test plan, security architecture, and documentation



PIONEERING IMPACT

Demonstrated that advanced AI experiences can be built and deployed fully on-device

FUTURE SCOPE & ROADMAP

01

PHASE 1 – FOUNDATION (0–3 MONTHS)

- Dockerize for one-click deployment
- Add GPU acceleration for XTTS
- Mobile-responsive PWA frontend
- Multi-language support (Hindi, Punjabi & more)



02

PHASE 2 – ENRICH & COLLABORATE (3–9 MONTHS)

- 3D avatar with lip-sync (Wav2Lip)
- Real-time streaming voice responses
- Multi-user collaboration
- Chrome extension for auto-posting



03

PHASE 3 – SCALE & EXPAND (9–15 MONTHS)

- SaaS launch with freemium model
- Enterprise API for businesses
- Wearable integration (AR glasses)
- Federated learning across devices



VISION

Alternate Ego as a universal personal AI layer — your personality, preserved and deployable anywhere, by anyone.

KEY IMPACT



Complete
Data Ownership



High Performance
on Local Devices



Personalized
AI Experience



Privacy,
Always



Future-Ready
Architecture



Thank You

We build immortality.



github.com/prabhs45667/new-alternate-ego



GTBIT, GGSIPU, New Delhi | CSE Batch 2022–26



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